

Data Analyst – Internship

Task 3: E-Commerce Database & Data Analysis

Tool: PostgreSQL


1 Database
Schema.sql

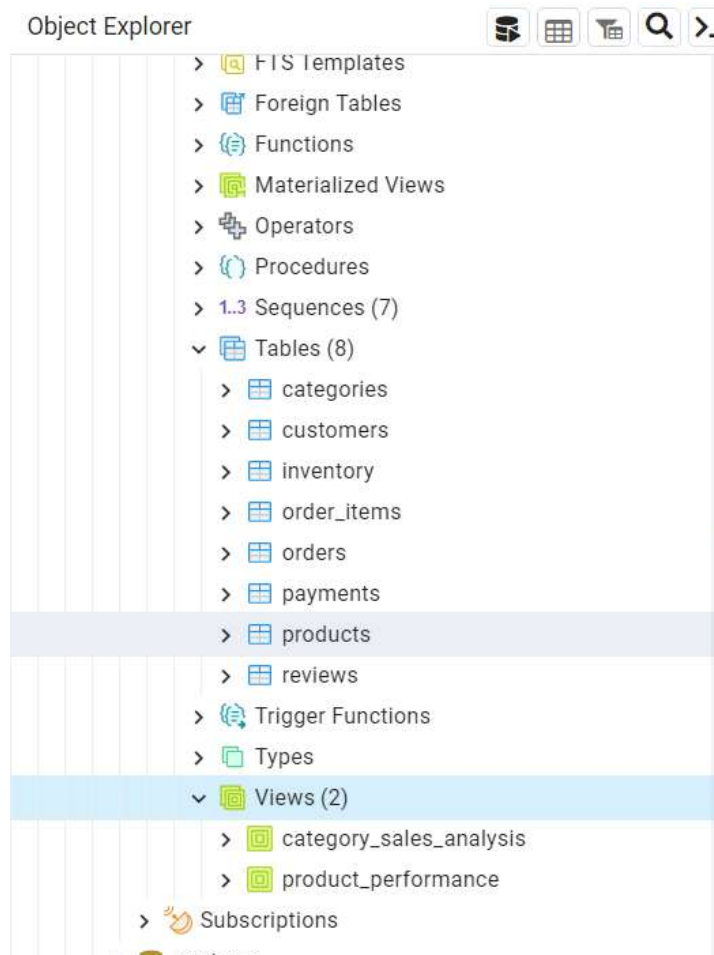

2 Mock Data
Scripts.sql


3 Indexes.sql


4 View - Category
Sales.sql


5 View - Product
Performance.sql

Database Schema:





6 Analysis SQL Queries.sql

Analysis SQL Query Outputs:

ecommerce/postgres@PostgreSQL 18*

Query Query History Scratch Pad x

```
1 --High Value Orders--
2
3 SELECT o.order_id, o.customer_id, CONCAT(c.first_name, ' ', c.last_name) AS customer_name
4 FROM orders o
5 INNER JOIN customers c ON o.customer_id = c.customer_id
6 WHERE total_amount > 500
7 ORDER BY total_amount DESC;
8
```

Data Output Messages Notifications

	order_id [PK] integer	customer_id integer	customer_name text	total_amount numeric (12,2)
1	5	1	John Doe	2599.00
2	10	2	Jane Smith	1299.50
3	1	1	John Doe	719.98
4	7	3	Michael Brown	699.99
5	15	3	Michael Brown	699.99
6	13	1	John Doe	549.98
7	3	3	Michael Brown	549.96

ecommerce/postgres@PostgreSQL 18*

Query Query History Scratch Pad x

```
9 --Products with high value--
10
11 SELECT product_id, name, price
12 FROM products
13 WHERE price > 500
14 ORDER BY price DESC;
15
```

Data Output Messages Notifications

	product_id [PK] integer	name character varying	price numeric (10,2)
1	2	Laptop	1299.50
2	1	Smartphone	699.99

ecommerce/postgres@PostgreSQL 18

Query Query History Scratch Pad

```

16 --Revenue per category--
17
18 SELECT
19     c.name AS category_name,
20     SUM(oi.quantity * oi.price) AS total_revenue
21 FROM categories c
22 INNER JOIN products p ON c.category_id = p.category_id
23 LEFT JOIN order_items oi ON p.product_id = oi.product_id
24 GROUP BY c.name
25 ORDER BY total_revenue DESC;
26

```

Data Output Messages Notifications

	category_name	total_revenue
1	Electronics	7948.39
2	Clothing	569.82
3	Books	119.94

ecommerce/postgres@PostgreSQL 18

Query Query History Scratch Pad

```

20
21 --Average rating per product--
22
23 SELECT
24     p.name,
25     COALESCE(ROUND(AVG(r.rating), 2), 0) AS avg_rating
26 FROM products p
27 LEFT JOIN reviews r ON p.product_id = r.product_id
28 GROUP BY p.name
29 ORDER BY avg_rating DESC NULLS LAST;
30

```

Data Output Messages Notifications

	name	avg_rating
1	Science Book	5.00
2	Smartphone	5.00
3	Jeans	5.00
4	T-Shirt	4.00
5	Tablet	4.00
6	Wireless Earbuds	3.00
7	Laptop	0
8	Novel	0

ecommerce/postgres@PostgreSQL 18

Query Query History Scratch Pad

```

37 --Top-selling products--
38
39 SELECT name
40 FROM products
41 WHERE product_id IN (
42     SELECT product_id
43     FROM order_items
44     GROUP BY product_id
45     ORDER BY SUM(quantity) DESC
46     LIMIT 2
47 );
48

```

Data Output Messages Notifications

	name
1	T-Shirt
2	Jeans

ecommerce/postgres@PostgreSQL 18

Query Query History Scratch Pad

```

48
49 --Most valuable customers--
50
51 SELECT
52     c.customer_id,
53     CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
54     SUM(o.total_amount) AS lifetime_value
55 FROM customers c
56 JOIN orders o ON c.customer_id = o.customer_id
57 GROUP BY c.customer_id, c.first_name
58 ORDER BY lifetime_value DESC;
59

```

Data Output Messages Notifications

	customer_id [PK] integer	customer_name text	lifetime_value numeric
1	1	John Doe	4268.95
2	3	Michael Brown	2099.93
3	2	Jane Smith	1799.41
4	4	Emily Davis	469.86

ecommerce/postgres@PostgreSQL 18

```

60 --Ratings and revenue--
61
62 SELECT
63     p.name,
64     COALESCE(SUM(oi.quantity * oi.price), 0) AS revenue,
65     COALESCE(ROUND(AVG(r.rating), 2), 0) AS avg_rating
66 FROM products p
67 LEFT JOIN order_items oi ON p.product_id = oi.product_id
68 LEFT JOIN reviews r ON p.product_id = r.product_id
69 GROUP BY p.name
70 ORDER BY avg_rating DESC, revenue DESC;
71

```

Data Output Messages Notifications

	name character varying	revenue numeric	avg_rating numeric
1	Smartphone	4698.97	5.00
2	Jeans	349.93	5.00
3	Science Book	89.97	5.00
4	Tablet	1199.97	4.00
5	T-Shirt	219.89	4.00
6	Wireless Earbuds	749.95	3.00
7	Laptop	1299.50	0
8	Novel	29.97	0

ecommerce/postgres@PostgreSQL 18

```

71
72 --Orders per product--
73
74 SELECT
75     p.product_id,
76     p.name AS product_name,
77     COALESCE(SUM(oi.quantity), 0) AS total_quantity_ordered
78 FROM products p
79 LEFT JOIN order_items oi ON p.product_id = oi.product_id
80 GROUP BY p.product_id, p.name
81 ORDER BY total_quantity_ordered DESC;
82

```

Data Output Messages Notifications

	product_id [PK] integer	product_name character varying	total_quantity_ordered bigint
1	3	T-Shirt	11
2	7	Jeans	7
3	1	Smartphone	5
4	6	Wireless Earbuds	5
5	8	Science Book	3
6	5	Tablet	3
7	4	Novel	3
8	2	Laptop	1

ecommerce/postgres@PostgreSQL 18

Query Query History Scratch Pad

```

83 --Revenue per product--
84
85 SELECT
86     p.product_id,
87     p.name AS product_name,
88     SUM(COALESCE(oi.quantity, 0)) AS total_quantity_ordered,
89     SUM(COALESCE(oi.quantity * oi.price, 0)) AS total_revenue
90 FROM products p
91 LEFT JOIN order_items oi ON p.product_id = oi.product_id
92 GROUP BY p.product_id, p.name
93 ORDER BY total_revenue DESC;
94

```

Data Output Messages Notifications

	product_id [PK] integer	product_name character varying	total_quantity_ordered bigint	total_revenue numeric
1	1	Smartphone	5	4698.97
2	2	Laptop	1	1299.50
3	5	Tablet	3	1199.97
4	6	Wireless Earbuds	5	749.95
5	7	Jeans	7	349.93
6	3	T-Shirt	11	219.89
7	8	Science Book	3	89.97
8	4	Novel	3	29.97

ecommerce/postgres@PostgreSQL 18

Query Query History Scratch Pad

```

94
95 --Orders per category--
96
97 SELECT
98     c.name AS category_name,
99     SUM(COALESCE(oi.quantity, 0)) AS total_quantity_ordered
100 FROM products p
101 INNER JOIN categories c ON p.category_id = c.category_id
102 LEFT JOIN order_items oi ON oi.product_id = p.product_id
103 GROUP BY c.name
104 ORDER BY total_quantity_ordered DESC;
105

```

Data Output Messages Notifications

	category_name character varying	total_quantity_ordered bigint
1	Clothing	18
2	Electronics	14
3	Books	6

ecommerce/postgres@PostgreSQL 18*

ecommerce/postgres@PostgreSQL 18

Query Query History Scratch Pad x

```

106 --Revenue per category--
107
108 SELECT
109     c.category_id,
110     c.name AS category_name,
111     COALESCE(SUM(oi.quantity * oi.price), 0) AS total_sales_value
112 FROM categories c
113 INNER JOIN products p ON c.category_id = p.category_id
114 LEFT JOIN order_items oi ON p.product_id = oi.product_id
115 GROUP BY c.category_id, c.name
116 ORDER BY total_sales_value DESC;
117

```

Data Output Messages Notifications

category_id	category_name	total_sales_value
[PK] integer	character varying	numeric
1	Electronics	7948.39
2	Clothing	569.82
3	Books	119.94

ecommerce/postgres@PostgreSQL 18*

ecommerce/postgres@PostgreSQL 18

Query Query History Scratch Pad x

```

118 --Product-level Review Analysis--
119
120 SELECT
121     p.product_id,
122     p.name AS product_name,
123     COUNT(r.review_id) AS total_reviews,
124     ROUND(AVG(r.rating), 2) AS avg_rating,
125     MAX(r.rating) AS highest_rating,
126     MIN(r.rating) AS lowest_rating
127 FROM products p
128 LEFT JOIN reviews r ON p.product_id = r.product_id
129 GROUP BY p.product_id, p.name
130 ORDER BY avg_rating DESC NULLS LAST;
131

```

Data Output Messages Notifications

product_id	product_name	total_reviews	avg_rating	highest_rating	lowest_rating
[PK] integer	character varying	bigint	numeric	integer	integer
1	Smartphone	1	5.00	5	5
2	Jeans	1	5.00	5	5
3	Science Book	1	5.00	5	5
4	T-Shirt	1	4.00	4	4
5	Tablet	1	4.00	4	4
6	Wireless Earbuds	1	3.00	3	3
7	Laptop	0	[null]	[null]	[null]
8	Novel	0	[null]	[null]	[null]

ecommerce/postgres@PostgreSQL 18

Query Query History Scratch Pad

```

132 --Products-wise Profit Analysis--
133
134 SELECT
135     p.product_id,
136     p.name AS product_name,
137     SUM(oi.quantity) AS total_quantity_sold,
138     SUM(oi.quantity * oi.price) AS total_revenue,
139     SUM(oi.quantity * p.cost_price) AS total_cost,
140     SUM(oi.quantity * oi.price) - SUM(oi.quantity * p.cost_price) AS total_profit
141 FROM products p
142 INNER JOIN order_items oi ON p.product_id = oi.product_id
143 GROUP BY p.product_id, p.name
144 ORDER BY total_profit DESC NULLS LAST;
145

```

Data Output Messages Notifications

	product_id [PK] integer	product_name character varying	total_quantity_sold bigint	total_revenue numeric	total_cost numeric	total_profit numeric
1	1	Smartphone	5	4698.97	2600.00	2098.97
2	5	Tablet	3	1199.97	840.00	359.97
3	2	Laptop	1	1299.50	1000.00	299.50
4	6	Wireless Earbuds	5	749.95	475.00	274.95
5	7	Jeans	7	349.93	175.00	174.93
6	3	T-Shirt	11	219.89	110.00	109.89
7	8	Science Book	3	89.97	36.00	53.97
8	4	Novel	3	29.97	13.50	16.47

ecommerce/postgres@PostgreSQL 18

Query Query History Scratch Pad

```

145
146 --Category-wise Profit Analysis--
147
148 SELECT
149     c.name AS category_name,
150     SUM(oi.quantity * oi.price) AS total_revenue,
151     SUM(oi.quantity * p.cost_price) AS total_cost,
152     SUM(oi.quantity * oi.price) - SUM(oi.quantity * p.cost_price) AS total_profit
153 FROM categories c
154 INNER JOIN products p ON c.category_id = p.category_id
155 INNER JOIN order_items oi ON p.product_id = oi.product_id
156 GROUP BY c.name
157 ORDER BY total_profit DESC;
158

```

Data Output Messages Notifications

	category_name character varying	total_revenue numeric	total_cost numeric	total_profit numeric
1	Electronics	7948.39	4915.00	3033.39
2	Clothing	569.82	285.00	284.82
3	Books	119.94	49.50	70.44

ecommerce/postgres@PostgreSQL 18

Query Query History Scratch Pad

```

159 --Profit Margin % by Product--
160
161 SELECT
162     p.name,
163     COALESCE(ROUND(
164         (SUM(oi.quantity * oi.price) - SUM(oi.quantity * p.cost_price))
165         / NULLIF(SUM(oi.quantity * oi.price),0) * 100,
166         2), 0) AS profit_margin_percentage
167 FROM products p
168 LEFT JOIN order_items oi ON p.product_id = oi.product_id
169 GROUP BY p.name
170 ORDER BY profit_margin_percentage DESC NULLS LAST;
171

```

Data Output Messages Notifications

	name character varying	profit_margin_percentage numeric
1	Science Book	59.99
2	Novel	54.95
3	Jeans	49.99
4	T-Shirt	49.97
5	Smartphone	44.67
6	Wireless Earbuds	36.66
7	Tablet	30.00
8	Laptop	23.05

ecommerce/postgres@PostgreSQL 18

Query Query History Scratch Pad

```

1 -- View: public.category_sales_analysis
2
3 -- DROP VIEW public.category_sales_analysis;
4
5 CREATE OR REPLACE VIEW public.category_sales_analysis
6 AS
7 SELECT c.category_id,
8        c.name AS category_name,
9        sum(oi.quantity::numeric * oi.price) AS total_revenue
10 FROM categories c
11 LEFT JOIN products p ON c.category_id = p.category_id
12 LEFT JOIN order_items oi ON p.product_id = oi.product_id
13 GROUP BY c.category_id, c.name;
14

```

Data Output Messages Notifications

	category_id integer	category_name character varying	total_revenue numeric
1	1	Electronics	7948.39
2	3	Books	119.94
3	2	Clothing	569.82

ecommerce/postgres@PostgreSQL 18

Query Query History Scratch Pad

```

4
5 CREATE OR REPLACE VIEW public.product_performance
6 AS
7 SELECT p.product_id,
8        p.name,
9        COALESCE(sum(oi.quantity), 0::bigint) AS total_sold,
10       COALESCE(sum(oi.quantity::numeric * oi.price), 0::numeric) AS total_revenue,
11       COALESCE(round(avg(r.rating), 2), 0::numeric) AS avg_rating
12 FROM products p
13 LEFT JOIN order_items oi ON p.product_id = oi.product_id
14 LEFT JOIN reviews r ON p.product_id = r.product_id
15 GROUP BY p.product_id, p.name
16 ORDER BY (COALESCE(sum(oi.quantity::numeric * oi.price), 0::numeric)) DESC;
17

```

Data Output Messages Notifications

	product_id integer	name character varying	total_sold bigint	total_revenue numeric	avg_rating numeric
1	1	Smartphone	5	4698.97	5.00
2	2	Laptop	1	1299.50	0
3	5	Tablet	3	1199.97	4.00
4	6	Wireless Earbuds	5	749.95	3.00
5	7	Jeans	7	349.93	5.00
6	3	T-Shirt	11	219.89	4.00
7	8	Science Book	3	89.97	5.00
8	4	Novel	3	29.97	0